Understanding the THREE Budget Lines
MC Problem 6

ECO 310

The Confusion

When we do substitution effect problems, there are **THREE budget lines**:

1. **Original budget line** (old prices, old income)
2. **New budget line** (new prices, same old income)
3. **Compensated budget line** (new prices, hypothetical higher income)

Let me show you exactly what each one is for MC6.

MC6 Setup

Initial situation:

- Prices: $p_B = \$1$ per beef stick, $p_T = \$4$ per toy
- Melsi buys: $(B, T) = (16, 4)$
- Income: $m = 1(16) + 4(4) = \$32$

After price change:

- New prices: $p_B = \$4$ per beef stick, $p_T = \$4$ per toy
- Income: still $m = \$32$ (no actual compensation!)

Budget Line #1: Original

$$\boxed{p_B \cdot B + p_T \cdot T = m}$$

Equation:

$$1 \cdot B + 4 \cdot T = 32$$

Rearranged:

$$T = 8 - \frac{B}{4}$$

Properties:

- **Slope:** $-\frac{1}{4}$ (relatively flat because beef is cheap)
- **B-intercept:** If $T = 0$, then $B = 32$ (can afford 32 beef sticks)
- **T-intercept:** If $B = 0$, then $T = 8$ (can afford 8 toys)
- **Optimal point:** $(16, 4)$ lies on this line

Check: $1(16) + 4(4) = 16 + 16 = 32$ ✓

What it means: This shows what Melsi could afford at the *old* prices with $32.

Budget Line #2: New

$$p_B^{\text{new}} \cdot B + p_T^{\text{new}} \cdot T = m$$

Equation:

$$4 \cdot B + 4 \cdot T = 32$$

Simplified:

$$B + T = 8$$

Rearranged:

$$T = 8 - B$$

Properties:

- **Slope:** -1 (steeper than original because beef is now more expensive relative to toys)
- **B-intercept:** If $T = 0$, then $B = 8$ (can only afford 8 beef sticks now)
- **T-intercept:** If $B = 0$, then $T = 8$ (can still afford 8 toys)
- **Optimal point:** $(4, 4)$ lies on this line

Check: $4(4) + 4(4) = 16 + 16 = 32$ ✓

What it means: This shows what Melsi can actually afford at the *new* prices with her same income of \$32.

Key observation: The budget line **rotated inward** around the T-intercept because the price of beef increased while the price of toys stayed the same.

Budget Line #3: Compensated

$$p_B^{\text{new}} \cdot B + p_T^{\text{new}} \cdot T = m^{\text{comp}}$$

The question: How much money would Melsi need at the *new* prices to achieve her *original utility* $u^* = 64$?

Finding the compensated income:

Original utility: $u^* = BT = 16 \times 4 = 64$

At new prices ($p_B = p_T = 4$), optimal allocation has $B = T$ (equal spending).

To achieve $u = 64$ with $B = T$:

$$B \cdot T = B \cdot B = B^2 = 64 \quad \Rightarrow \quad B = 8, T = 8$$

Cost: $m^{\text{comp}} = 4(8) + 4(8) = 64$

Equation:

$$4 \cdot B + 4 \cdot T = 64$$

Simplified:

$$B + T = 16$$

Rearranged:

$$T = 16 - B$$

Properties:

- **Slope:** -1 (same as new budget line!)
- **B-intercept:** If $T = 0$, then $B = 16$
- **T-intercept:** If $B = 0$, then $T = 16$
- **Optimal point:** $(8, 8)$ lies on this line

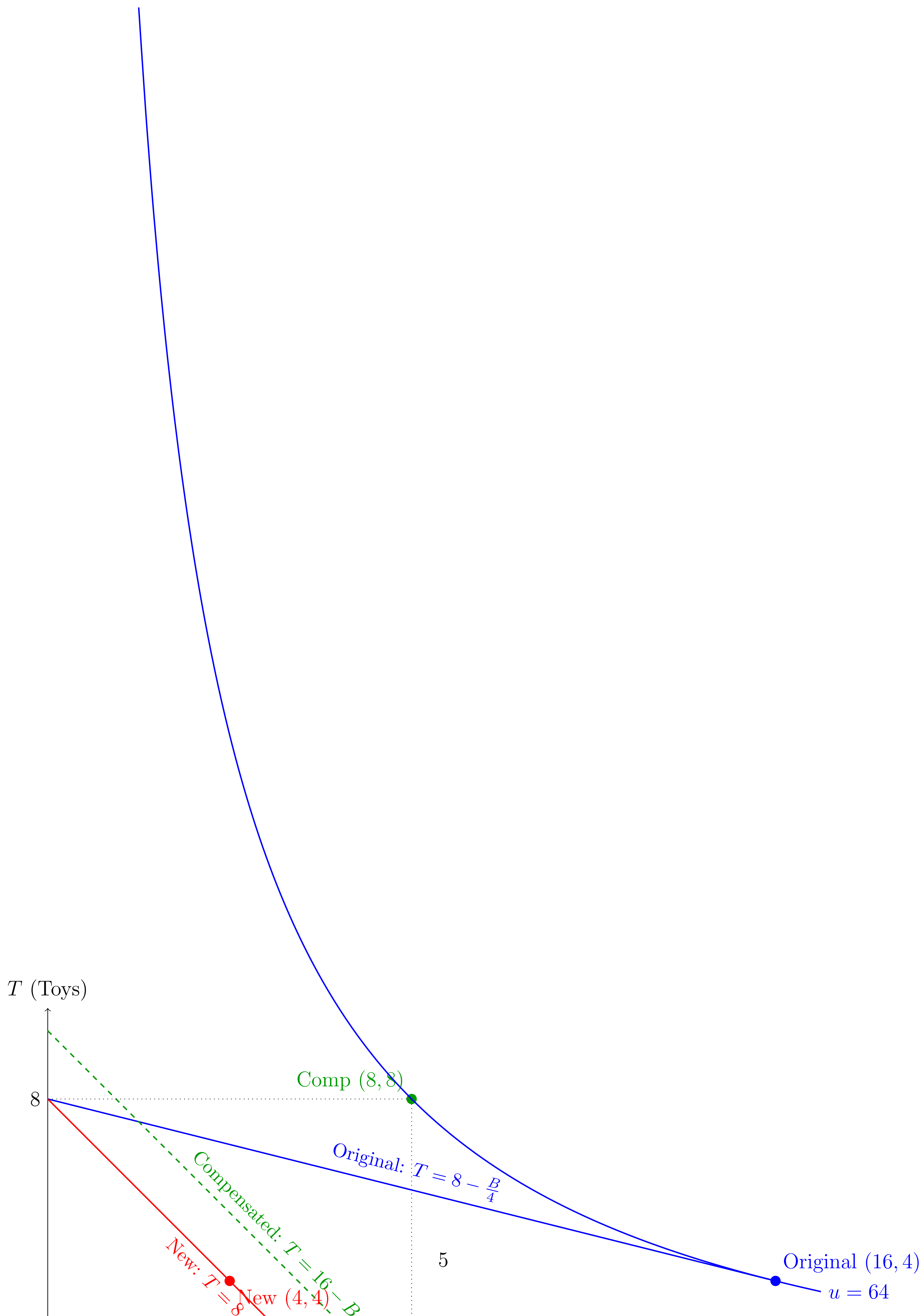
Check: $4(8) + 4(8) = 32 + 32 = 64$ ✓

And utility: $u = 8 \times 8 = 64$ ✓

What it means: This is **hypothetical**. It shows what Melsi could afford if we gave her \$64 instead of \$32 to compensate her for the price increase.

Key observation: The compensated budget line is **parallel** to the new budget line (both have slope -1) but shifted outward because she has more money (\$64 vs \$32).

Visual Comparison



T (Toys)
8
Comp $(8, 8)$
Original: $T = 8 - \frac{B}{4}$
Compensated: $T = 16 - B$
New: $T = 8$
5
New $(4, 4)$
Original $(16, 4)$
$u = 64$

Key observations:

1. **Blue line (original)** is flatter because beef was cheap
2. **Red line (new)** rotated inward and got steeper because beef got expensive
3. **Green dashed line (compensated)** is parallel to red but shifted out (more money)
4. All three have the *same T-intercept = 8* because toy price didn't change... **WAIT, NO!**

CORRECTION: Let me recalculate the intercepts properly!

Intercepts - The Right Way

Original Budget Line: $B + 4T = 32$

- B-intercept ($T = 0$): $B = 32$
- T-intercept ($B = 0$): $4T = 32 \Rightarrow T = 8$

New Budget Line: $4B + 4T = 32$

- B-intercept ($T = 0$): $4B = 32 \Rightarrow B = 8$
- T-intercept ($B = 0$): $4T = 32 \Rightarrow T = 8$

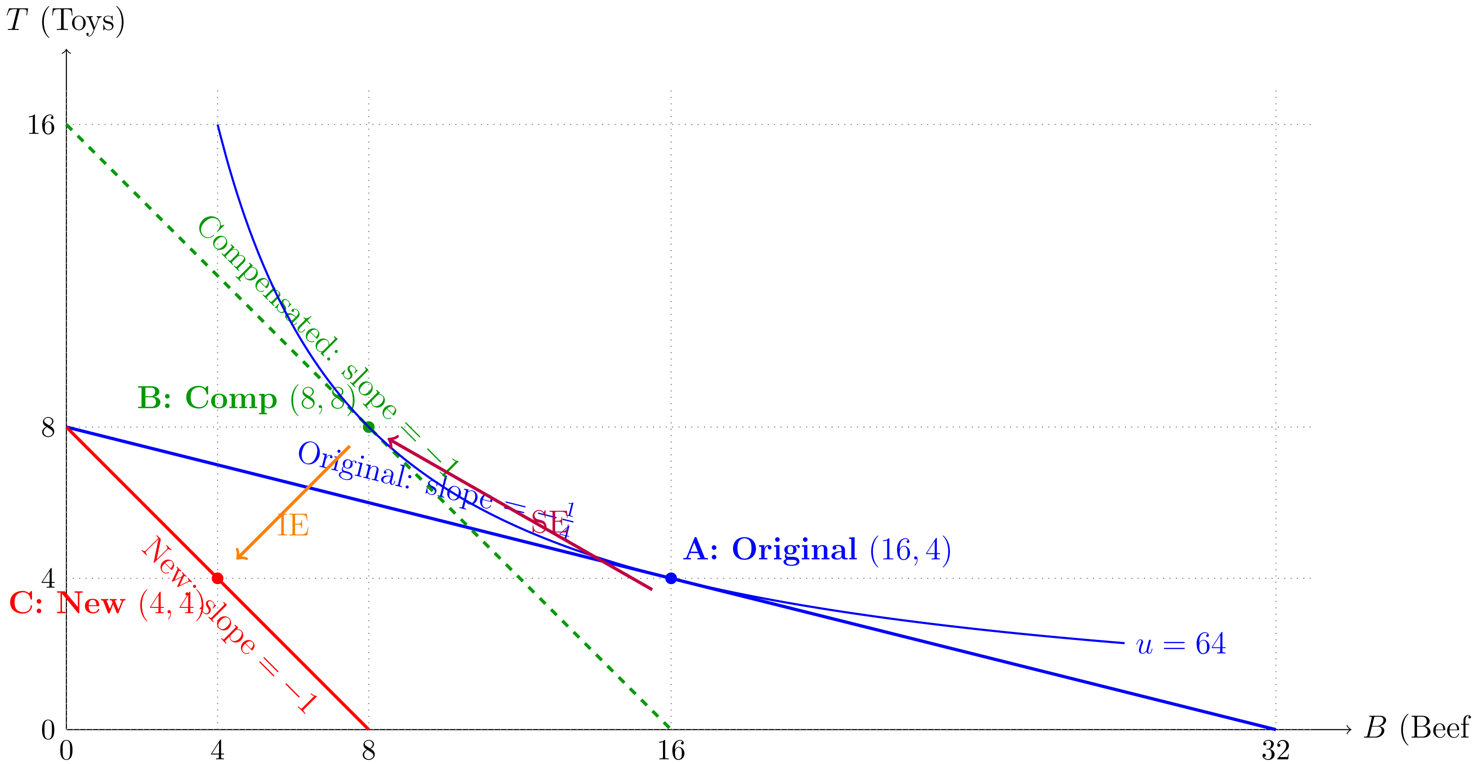
Notice: Both the original and new lines have T-intercept = 8! This makes sense because:

- If Melsi spends all \$32 on toys at old prices: $32/4 = 8$ toys
- If Melsi spends all \$32 on toys at new prices: $32/4 = 8$ toys
- Toy price didn't change, so T-intercept stays the same!

Compensated Budget Line: $4B + 4T = 64$

- B-intercept ($T = 0$): $4B = 64 \Rightarrow B = 16$
- T-intercept ($B = 0$): $4T = 64 \Rightarrow T = 16$

Better Visual with Correct Intercepts



T (Toys)
16
8
4
0
0
4
8
16
32
B (Beef
Compensated: slope $= -1$
B: Comp $(8, 8)$
Original: slope $= -\frac{1}{4}$
SE
IE
A: Original $(16, 4)$
C: New $(4, 4)$
New: slope $= -1$
$u = 64$

Why Each Line is Where It Is

1. Why is the original line flatter?

Slope $= -\frac{p_B}{p_T} = -\frac{1}{4}$

When beef costs \$1 and toys cost \$4, you only give up $\frac{1}{4}$ of a toy to get 1 beef stick. The line is relatively flat.

2. Why does the new line rotate inward?

Slope $= -\frac{p_B}{p_T} = -\frac{4}{4} = -1$

When beef costs \$4 and toys cost \$4, you give up exactly 1 toy to get 1 beef stick. The line gets steeper.

Why inward? Because beef got more expensive, you can afford less beef. The B-intercept shrinks from 32 to 8.

Why same T-intercept? Toy price didn't change, so if you spend all money on toys, you can still buy 8 toys.

3. Why is the compensated line parallel to the new line?

Same slope $= -1$ because it uses the same new prices ($p_B = p_T = 4$).

But shifted out because we give Melsi \$64 instead of \$32, so she can afford more of everything.

4. Why does the compensated line pass through $(8, 8)$?

Because that's the cheapest bundle at new prices that achieves $u = 64$.

Summary Table

Budget Line	Prices	Income	Slope	Optimal Point
Original	$(1, 4)$	\$32	$-\frac{1}{4}$	$(16, 4)$
New	$(4, 4)$	\$32	-1	$(4, 4)$
Compensated	$(4, 4)$	\$64	-1	$(8, 8)$

The Key Insight

The compensated budget line is NOT real!
It's a hypothetical line showing: "What if we gave Melsi enough money to stay at $u = 64$ after the price change?"
In reality, Melsi goes from the Original line to the New line. The Compensated line is just a tool to separate SE and IE.

Common Mistakes

1. **Thinking all three lines pass through the same point** - They don't! Each has a different optimal bundle.
2. **Confusing which line uses which income** - Original and New use \$32; Compensated uses \$64.
3. **Thinking the compensated line uses old prices** - No! It uses new prices with more income.
4. **Not understanding why slopes change** - Slope $= -\frac{p_1}{p_2}$, so when relative prices change, slope changes.